Deployment & Customer Handover Guide

DrowsyGuard DMS Production Operations & Maintenance Manual | MASTER_SDLC_AGENTS

1. Executive Handover Overview

This document certifies formal technical delivery and operational handover of the Real-Time AI Driver Drowsiness, Fatigue & Microsleep Detection System. The deliverable encompasses full source code (src/), automated regression test suites (tests/), Docker containerization manifests, and security certification reports.

2. System Requirements & Hardware Specifications

- **Operating System:** Ubuntu 20.04+, Debian 11+, Windows 10/11, macOS 12+ (x86_64 or Apple Silicon ARM64).
- **Compute Specifications:** Dual-Core 2.0 GHz x86/ARM CPU (e.g. Raspberry Pi 4/5, Intel Core i3+, NVIDIA Jetson).
- **Memory & Storage:** 2 GB RAM minimum, 500 MB free disk space for container image and assets.
- **Camera Ingestion:** Standard 720p 30 FPS USB webcam or vehicle cabin RTSP near-infrared (NIR) video stream.

3. Installation & Local Operational Launch

- **Step 1 (Environment):** python -m venv venv && source venv/bin/activate (or .\venv\Scripts\activate on Windows).
- **Step 2 (Dependencies):** pip install -r requirements.txt
- **Step 3 (Launch Console):** streamlit run src/app.py --server.port=8501

4. Interactive Cockpit Console Operations

- **Preset Driving Demos:** Switch between real in-cabin driver benchmarks (Attentive, Yawn, Microsleep) via dropdown.
- **Live Webcam Stream:** Take camera snapshots or stream cabin video to verify real-time driver telemetry.
- **Upload Custom Media:** Drag and drop any automotive cabin video (.mp4) or photo (.jpg/.png) for instant analysis.
- **1-Click CSV Log Export:** Download the complete frame-by-frame RFC 4180 audit log directly from the cockpit.

5. Automated Regression Verification & Maintenance

Run automated PyTest regression testing anytime before customer fleet deployment:

```
pytest -v tests/
```

Expected Pass Rate: 35 / 35 tests passed (100.0% verified).